

(42) notes I Prigda de

$$H = \frac{p^2}{2m} + \frac{1}{2} m \omega_0^2 x^2$$

→ change of var.

$$q = \sqrt{\frac{m\omega}{2}} \left(x + \frac{ip}{m\omega} \right), \quad q^\dagger = \sqrt{\frac{m\omega}{2}} \left(x - \frac{ip}{m\omega} \right)$$

$$\rightarrow [q, q^\dagger] = 1$$

$$\rightarrow H = \omega \left(a^\dagger a + \frac{1}{2} \right)$$

changing eig. are
eig. of $\hat{N} = a^\dagger a$

$$\hat{N} |u\rangle = u |u\rangle$$

$$a^\dagger |u\rangle = \sqrt{u+1} |u+1\rangle$$

$$a |u\rangle = \sqrt{u} |u-1\rangle$$

Heisenberg pic.

$$i \frac{d}{dt} a = [a, H] = \left[a, \omega \left(a^\dagger a + \frac{1}{2} \right) \right] = \omega (a a^\dagger a - a^\dagger a a)$$

$$= -i \omega [a, a^\dagger] a = -i \omega a$$

$$\rightarrow a(t) = e^{-i\omega t} a(0)$$

S.P.:

$\mathcal{D}d = e$ (free field) *

$$\phi(\vec{r}, t) = \int \frac{d^3p}{(2\pi)^3} \left[a_p(t) e^{i\vec{p}\vec{r}} + a_p^\dagger(t) e^{-i\vec{p}\vec{r}} \right]$$

2nd quant.

free Ham. / bosons

$$H_0 = \int \frac{d^3p}{(2\pi)^3} \omega_p (a_p^\dagger a_p + 1/2)$$

$$\omega_p = |\vec{p}|$$

- 1 GM system per $\vec{p}$

- the excitations of $\vec{p}$ h.c. boson particles

- 2nd quant. $\mathcal{U} \rightarrow \mathcal{F} \equiv \bigoplus_n \mathcal{U}_n$

hilbert $\rightarrow$ Fock

eg. q. states are $|p_1^m \dots p_n^m\rangle$

Field expansion

$$[a_n, a_p^\dagger] = 1$$

is now

$$[a_n, a_p^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{n})$$

$$a_1^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_1}} |\vec{p}\rangle$$

$$\langle 0|0\rangle = 1$$

$$A \langle \vec{p} | \vec{k} \rangle = 2\sqrt{\omega_p \omega_k} \langle 0 | a_p a_k^\dagger | 0 \rangle = 2\omega_p (2\pi)^3 \delta^3(\vec{p} - \vec{k})$$

identity of:

$$\mathbb{I} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |\vec{p}\rangle \langle \vec{p}|$$

Free field:

$$\phi_0(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}})$$

$$\langle \vec{p} | \phi_0(\vec{x}) | 0 \rangle = 2\omega_p \sqrt{\omega_p} a_p \int \frac{d^3k}{(2\pi)^3} (a_k e^{i\vec{k}\vec{x}} + a_k^\dagger e^{-i\vec{k}\vec{x}}) | 0 \rangle$$

$$= \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_p}{\omega_k}} [e^{i\vec{k}\vec{x}} \langle 0 | a_p a_k | 0 \rangle + e^{-i\vec{k}\vec{x}} \langle 0 | a_p a_k^\dagger | 0 \rangle]$$

$$= e^{-i\vec{p}\vec{x}}$$

$$\Rightarrow \phi_0(\vec{x}) | 0 \rangle = | \vec{x} \rangle$$

Time deji:

$$a_p^\dagger(t) = e^{i\omega_p t} a_p^\dagger, \quad a_p = e^{-i\omega_p t} a_p$$

$$\text{I define } \phi(\vec{r}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx})$$

$$\begin{aligned} [H_0, \phi(\vec{r}, t)] &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{\omega_k}} \left[\omega_p (a_p^\dagger a_p + \frac{1}{2}) \right. \\ &\quad \left. a_k e^{-ikx} + a_k^\dagger e^{ikx} \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{\omega_p}} \left[-\omega_p a_p e^{-ipx} + \omega_p a_p^\dagger e^{ipx} \right] \\ &= -i \partial_t \phi(\vec{r}, t) \end{aligned}$$

Heisenberg Eqn:

$$i \partial_t \phi(\vec{r}, t) = [\phi, H]$$

$$\rightarrow \phi(\vec{r}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{\omega_p}} \left[a_p(t) e^{-ipx} + a_p^\dagger(t) e^{ipx} \right]$$